



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

NATIONAL INSTITUTE OF TECHNOLOGY ROURKELA

Course Details

CS3003 COMPUTER NETWORKS	L-T-P: 3-0-0, Credits: 3
SUBJECT NATURE	THEORY
FACULTY INSTRUCTOR	DR. SANJEEV PATEL

COURSE OBJECTIVES

- The student should understand the basic principles and fundamental design issues of computer networks.
- The student should understand the OSI layer and learn to distinguish the problems involved at each layer.
- To learn the HTTP, DNS, SMTP, and FTP protocols at the application layer.
- To understand and learn the transport layer protocols and their significance for achieving reliable data transfer.
- To design the IP addressing problems and practice the various routing problems.
- To learn the link layer protocols and design issues of the MAC layer.

COURSE OUTCOMES

The course aims to provide knowledge of networking functionalities operated between the computers. After completing this theory course, a student will be able to:

- **CO1:** Define the networking basics, TCP/IP protocol suite and OSI Reference model.
- **CO2:** Illustrate the various application layer protocols.
- **CO3:** Examine various transport protocols and their performance-enhancing mechanisms.
- **CO4:** Determine the shortest path for the network using various routing protocols and design the IP addressing.
- **CO5:** Apply the data link layer protocol to solve communication, error detection and correction problems

DESCRIPTION

This course will cover various protocols of the TCP/IP protocol suite used on the Internet by illustrating the OSI reference model.

LECTURE PLAN / SYLLABUS

Sr. No.	Course Contents	Lecture Plan
1	Network fundamentals, protocols and standards, reference models, the significance of layered network architectures, packet switching and circuit switching, types of transmissions, IEEE 802.3 Ethernet cables for Physical layer, types of networks	L1-L6
2	Application layer protocols: Client-server model, DNS, HTTP, FTP, SMTP, TELNET	L7-L12
3	Congestion, transport layer protocols, UDP, TCP, reliable transport connections,	L13-L18
4	Network layer protocols, IP addressing, Sub-netting and super-netting, Routing Algorithms, Internet Protocol (IP), ARP, RARP	L19-L26
5	MAC Layer protocols, contention-based protocols vs contention-free MAC protocols, Framing, CSMA/CD	L27-L32
6	Link layer aspects, synchronous and asynchronous transmission, Data link layer services, flow control, error control, ARQ, stop-n-wait protocols, Sliding window protocols; Wireless networks: wireless MAC protocols	L33-L40

Text Books:

1. *Computer Networking: A Top-Down Approach Featuring the Internet* by James Kurose and Keith Ross, Addison Wesley

References:

1. *Computer Networks* by Andrew S. Tanenbaum, Prentice-Hall Publishers
2. *Computer Networks a Systems Approach* by Larry Peterson and Bruce Davie, Morgan Kaufmann
3. *Data and Computer Communications* by William Stallings, Prentice Hall
4. *Data Communication and Networking* by Behrouz A Forouzan, Tata McGraw-Hill
5. K. Thramboulidis, A. Mikroyannidis, "Using UML for the Design of Communication Protocols: The TCP Case Study" 11th International Conference on Software, Telecommunications and Computer Networks, October 7-10, 2003.
6. Juha Parssinen, Niklas von Knorring, Jukka Heinonen, Markku Turunen, "UML for Protocol Engineering- Extensions and Experiences", *Proceedings of the Technology of Object-Oriented Languages and Systems (TOOLS)*,. IEEE Computer Society, . page 82, 2000.

Journals:

1. Transactions on Networking (ACM)
2. IEEE Transactions on Networking
3. IEEE Transactions on Distributed and Parallel Processing